5th Multi-modal Aerial View Imagery Challenge: Classification - PBVS 2026

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Abstract

This paper reports the results of the 5th Multi-modal Aerial View Imagery Challenge: Classification (MAVIC-C), which focuses on SAR-based aerial target classification and out-of-distribution (OOD) detection. The challenge is conducted on the UNICORN V2 dataset, which provides spatially aligned EO and SAR imagery to enable the study of cross-modal learning strategies for improving SAR recognition. Performance is evaluated using Top-1 Accuracy (Acc@1) for in-distribution classification and AUC together with TNR@TPR95 for OOD detection, with the final ranking determined by a weighted score prioritizing classification accuracy. The submitted methods largely rely on cross-modal feature learning, typically through dual-stream architectures that transfer structural information from EO to SAR representations. In the 2026 edition, 63 teams participated and 606 submissions were made. Among the evaluated approaches, AeroVision achieves the highest overall score (0.4176) due to superior classification performance, driven by its cross-modal domain adaptation framework that combines dual-stream EO-SAR feature learning with an auxiliary SAR encoder and collaborative optimization. In contrast, the DINOv3-based approach attains the best OOD detection with a TNR@TPR95 of 0.2695 by explicitly decoupling classification and uncertainty estimation.

1. Introduction

Aerial image classification in remote sensing refers to the automatic identification of objects [7, 12, 28], land-use categories [6], or targets [14] from imagery captured at high altitudes by airborne or satellite platforms [6]. Aerial image classification plays a crucial role in applications such as environmental monitoring [31], urban development analysis [36], disaster management [25], agricultural assessment [4], and defence operations [22], where timely and accurate interpretation of aerial data is essential for informed decision-making.

Unlike ground-level vision tasks, aerial image analysis must contend with large scene variability, scale ambiguity [37], limited spatial resolution, and viewpoint-induced distortions [5, 20]. Objects of interest often appear small [35], densely distributed, or partially occluded within complex backgrounds [5, 7], making robust classification particularly challenging. For these reasons, Deep Learning methods have become a dominant approach in aerial image classification [20]. DL models are trained on large amounts of annotated data [27], allowing them to learn representative textures, spatial patterns, and structural features that enable reliable and fully automated scene categorization.

Electro Optical (EO) imagery is typically the most widely used modality in this domain due to its ease of acquisition and intuitive interpretability [9]. However, it presents critical limitations, including its complete dependence on solar illumination and its inability to penetrate cloud cover [1], smoke, or adverse atmospheric conditions [29], which restricts reliable and continuous monitoring during nighttime or inclement weather [1, 29]. These constraints complicate the deployment of robust aerial recognition systems in real-world operational scenarios.

In response, alternative sensing modalities have gained increasing attention. In particular, Synthetic Aperture Radar (SAR) imaging provides a powerful complementary solution. Unlike optical sensors, SAR [2] does not capture color information [29] but instead measures the physical and dielectric properties [1] of targets through the interaction of microwave signals with the scene [2], encoding structural patterns and surface textures. SAR operates independently of solar illumination, enabling consistent imaging during nighttime and the ability to penetrate clouds, smoke, and light fog [3].

Nevertheless, SAR image classification remains a non-trivial task. SAR lacks color cues and is inherently affected by speckle noise [26], which complicates visual interpretation and feature extraction [23, 26]. The backscattered signal depends strongly on target geometry, material properties, surface roughness, and orientation relative to the sensor, leading to significant intra-class variability and complex scattering patterns [38]. Furthermore, SAR data generally offer lower semantic interpretability compared to optical imagery [8], and the availability of large-scale, well-annotated datasets remains limited [24, 34]. As a result, SAR-based aerial classification remains an active and relevant area of research.

This document presents the report of the 5th Multi-modal Aerial View Imagery Challenge: Classification (MAVIC-C). This edition continues the series of previous successful challenges [13, 17, 19] focused on advancing Automated Target Recognition (ATR) in aerial imagery. The classification track concentrates on SAR image classification, where the objective is to develop modern learning-based approaches to accurately classify SAR image chips across ten target categories and to identify Out-of-Distribution (OOD) samples. In addition to evaluating predictive performance, the challenge is designed to investigate whether complementary information from EO imagery can enhance the learning process for SAR-based recognition. This report summarizes the methodologies and results obtained by the top-three teams and analyzes the insights derived from the proposed approaches. In this edition, 63 teams participated and 606 submissions were made.

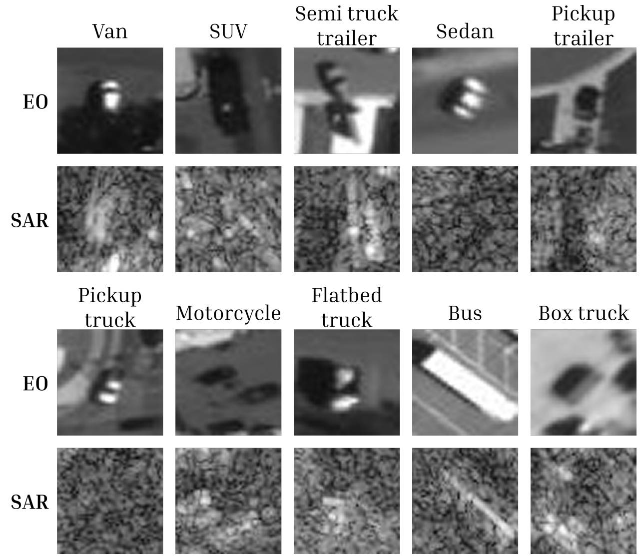

Figure 1. Representative examples of EO and SAR pairs from the UNICORN V2 dataset used in MAVIC-C.

2. Challenge description

2.1. Overview

The 5th Multi-modal Aerial View Imagery Challenge: Classification focuses on advancing automatic target recognition in SAR imagery. The primary objective of the challenge is to develop learning-based models capable of accurately classifying SAR image chips into ten predefined target categories while also identifying OOD samples. The challenge is designed to evaluate the robustness, generalization capability, and adaptability of modern classification approaches under realistic SAR imaging conditions. A dedicated dataset was provided for this purpose, enabling a standardized and controlled benchmarking framework. Through this setup, MAVIC-C aims to foster methodological advances in SAR-based aerial image classification and to promote reproducible evaluation within the research community.

2.2. Dataset

The MAVIC-C challenge is built upon the UNIfied COincident Optical and Radar for recognitioN (UNICORN V2) dataset [18], a multi-sensor aerial imagery collection designed for target recognition research. The original UNICORN[1] dataset was introduced in 2008 and comprises co-registered EO and Synthetic SAR data acquired over Dayton, Ohio, using large-format airborne sensors. The dataset includes Wide Area Motion Imagery (WAMI) EO data alongside wide-area SAR imagery.

To ensure precise spatial correspondence between modalities, the EO and SAR data were carefully aligned using geo-registration and homography-based refinement techniques. Target instances were manually annotated by human experts and subsequently extracted into image chips for classification purposes.

UNICORN V2 extends this resource and provides a structured benchmark tailored for classification and OOD evaluation. The dataset contains 20 object categories in total. The 10 most prevalent categories form the In-distribution Data (IDD) set used for supervised training and closed-set evaluation, while the remaining 10 less frequent categories are designated as OOD classes and serve as confusers for robustness assessment. The distribution of IDD classes and their respective splits are summarized in Table 1, while the OOD categories used for robustness evaluation are detailed in Table 2. Also, examples of EO and SAR pairs from the dataset are shown in Fig. 1.

2.3. Evaluation protocol

Challenge submissions are evaluated by comparing model predictions against reference ground-truth labels for IDD classification and by assessing the ability to separate OOD samples from IDD samples. Final ranking is computed on a sequestered test set of 4,000 SAR patches (56×56), consisting of 2,000 IDD samples (200 examples per each of the 10 IDD classes) and 2,000 OOD samples. Model performance is measured using the metrics described below.

[1] https://github.com/AFRL-RY/data-unicorn-2008

Table 1. In-distribution classes derived from UNICORN V2.

Vehicle Type	#Train	#Val	#Test
Sedan	364,291	77	200
SUV	43,401	77	200
Pickup truck	24,158	77	200
Van	16,890	77	200
Box truck	2,896	77	200
Motorcycle	1,441	77	200
Flatbed truck	898	77	200
Bus	612	77	200
Pickup truck w/ trailer	695	77	200
Semi truck w/ trailer	353	77	200
Total	**455,635**	**770**	**2000**

Table 2. Out-of-distribution classes used in UNICORN V2.

Vehicle Type	#Val	#Test
Van w/ trailer	77	–
Other	77	2000
Dismount	77	–
Semi	7	–
SUV w/ trailer	77	–
Flatbed truck w/ trailer	77	–
Plane	77	–
Bicycle	24	–
Dump truck	77	–
Sedan w/ trailer	77	–
Total	**647**	**2000**

Top-1 Accuracy. IDD classification performance is measured using standard Top-1 Accuracy (Acc@1) defined as in Eq. (1):

$$\mathrm{Acc@1} = \frac{1}{N_{\mathrm{IDD}}} \sum_{i=1}^{N_{\mathrm{IDD}}} \mathbb{I}\left[\hat{y}_i = y_i\right], \tag{1}$$

where N_{IDD} denotes the number of IDD test samples, y_i is the ground-truth class label, $\hat{y}_i$ is the predicted label, and $\mathbb{I}[\cdot]$ is the indicator function. Higher values of Acc@1 indicate better classification performance, with 1 corresponding to perfect agreement between predicted and ground-truth labels across all IDD test samples.

OOD detection and scoring function. OOD detection is evaluated using a scalar *OOD score* $s(x)$ produced by the model for each input chip x, where higher values indicate greater confidence that the sample is IDD (any monotone convention is acceptable as long as it is applied consistently when computing the ROC curve). Given a decision threshold τ, predictions are defined as IDD if $s(x) \geq \tau$ and OOD otherwise. The True Positive Rate (TPR) and False Positive Rate (FPR) at threshold τ are defined as in Eqs. (2) and (3):

$$\mathrm{TPR}(\tau) = \frac{\mathrm{TP}(\tau)}{\mathrm{TP}(\tau) + \mathrm{FN}(\tau)}; \tag{2}$$

$$\mathrm{FPR}(\tau) = \frac{\mathrm{FP}(\tau)}{\mathrm{FP}(\tau) + \mathrm{TN}(\tau)}, \tag{3}$$

where TP and FN denote correctly and incorrectly detected OOD samples, respectively, and FP and TN denote IDD samples that are incorrectly or correctly classified as OOD.

AUC. We report the Area Under the Receiver Operating Characteristic Curve (AUC), which summarizes separability between IDD and OOD across all thresholds, as shown in Eq. (4):

$$\mathrm{AUC} = \int_0^1 \mathrm{TPR}(\mathrm{FPR}) \; d(\mathrm{FPR}), \tag{4}$$

with $\mathrm{AUC} \in [0, 1]$, where values closer to 1 indicate improved discrimination (high TPR at low FPR).

TNR@TPR95. In addition to AUC, we report the True Negative Rate at 95% True Positive Rate (TNR@TPR95), a commonly used operating-point metric in OOD detection. The TNR is defined as shown in Eq. (5):

$$\mathrm{TNR}(\tau) = \frac{\mathrm{TN}(\tau)}{\mathrm{TN}(\tau) + \mathrm{FP}(\tau)} = 1 - \mathrm{FPR}(\tau). \tag{5}$$

To compute TNR@TPR95, we first determine the threshold $\tau^\star$ such that $\mathrm{TPR}(\tau^\star) = 0.95$ (or the closest achievable value). The reported metric is then given by Eq. (6):

$$\mathrm{TNR@TPR95} = \mathrm{TNR}(\tau^\star). \tag{6}$$

Final score. Consisted with previous editions, the final challenge score is computed as a weighted combination of IDD classification accuracy and OOD detection AUC, as defined in Eq. (7):

$$\text{Final score} = 0.75 \cdot \mathrm{Acc@1} + 0.25 \cdot \mathrm{AUC}. \tag{7}$$

This weighting prioritizes accurate target recognition on known classes while still rewarding models that reliably detect unknown targets.

3. Proposed strategies

3.1. AeroVision

The team **AeroVision** proposes a cross-modal semi-supervised domain adaptation framework improve SAR-based aerial target classification by exploiting complementary information from aligned EO imagery, shown in Fig. 2. The method focuses on learning transferable representations across heterogeneous sensing modalities while maintaining reliable OOD detection. To achieve this, the framework integrates cross-modal feature alignment, contrastive learning, and auxiliary model collaboration within a unified training pipeline.

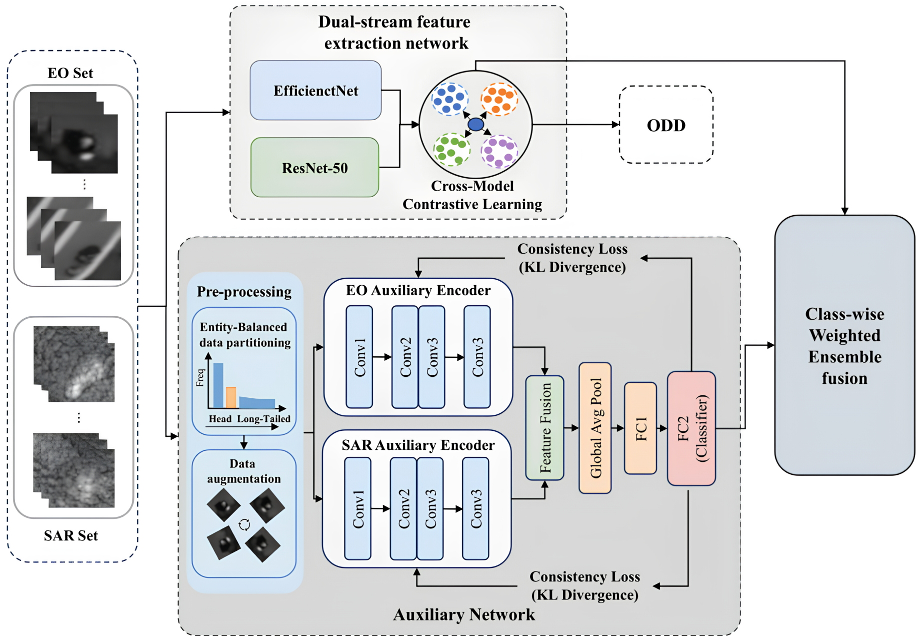

Figure 2. Cross-modal semi-supervised domain adaptation framework propsoed by AeroVision team.

3.1.1. Dual-stream cross-modal framework

The architecture adopts a dual-stream feature extraction network in which EO and SAR inputs are processed by separate backbones. Specifically, EfficientNet [32] is used to extract features from EO images, while ResNet50 [10] is employed for SAR imagery. These two branches learn complementary representations and are trained with domain adaptation objectives to align the feature spaces across modalities.

3.1.2. Cross-modal contrastive domain adaptation

To improve representation transfer between EO and SAR modalities, the framework introduces a cross-modal contrastive domain adaptation module. This component aims to learn domain-invariant features by encouraging alignment between EO and SAR representations while preserving class-level separability. The module integrates contrastive learning with metric-based category supervision to guide the feature extractor toward representations that remain discriminative across heterogeneous sensing modalities.

3.1.3. Auxiliary SAR encoder

To further enhance IDD classification performance, the framework introduces SimpleEncoder, which is a lightweight auxiliary model. This auxiliary network processes SAR imagery through three convolutional blocks with progressively increasing channels, followed by multi-scale feature fusion and a lightweight classification head. After feature fusion, MixStyle is applied to improve domain generalization. The auxiliary encoder is trained jointly with the main SAR branch through knowledge distillation and consistency constraints, encouraging the auxiliary features and predictions to remain aligned with those of the main model.

3.1.4. Knowledge distillation and consistency learning

The auxiliary encoder is guided by the main SAR branch through feature-level distillation and prediction-level consistency learning. Feature alignment between the auxiliary and main branches is encouraged through a Maximum Mean Discrepancy (MMD) objective applied to intermediate representations. In addition, a symmetric Kullback-Leibler (KL) divergence is applied between the logits of the auxiliary and main classifiers to enforce prediction consistency between both models.

3.1.5. Entity-balanced sampling

To address the severe class imbalance and the strong correlation between samples belonging to the same object instance, the training procedure adopts an entity-balanced sampling strategy. For each batch, a class is first selected, followed by the selection of an entity (cluster) within that class, from which a sample is drawn. This strategy increases diversity across entities within each batch and mitigates the long-tail distribution of the dataset, improving generalization for minority classes.

3.1.6. Joint optimization objective

The overall training objective combines several complementary loss terms. Classification for both the main SAR

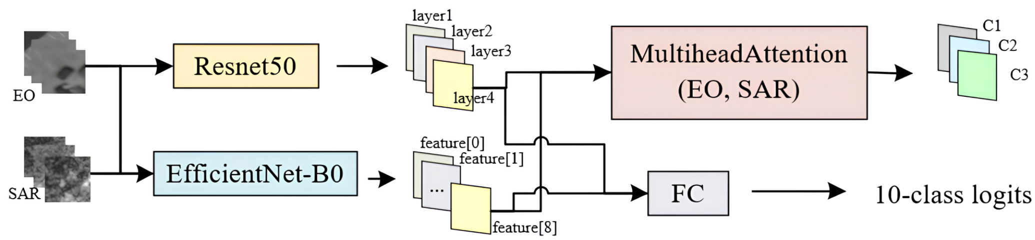

Figure 3. ResEffiNet model proposed by Songyq team.

branch and the auxiliary SimpleEncoder is optimized using a Focal Loss [16] to emphasize difficult samples under class imbalance. Cross-modal alignment between EO and SAR representations is encouraged through a Sliced Wasserstein Distance [33], which reduces the distribution discrepancy between the two modalities. In addition, knowledge distillation between the auxiliary encoder and the main SAR branch is enforced using a MMD loss applied to intermediate feature statistics. Finally, prediction consistency between both models is encouraged through a symmetric KL divergence between their output logits. The final objective is formulated as a weighted combination of these components.

3.1.7. Hardware and software specifications

- **Hardware:** Experiments were conducted on two NVIDIA RTX 3090 GPUs (24 GB VRAM each) with 32 GB of system RAM, running Ubuntu 22.04.
- **Software:** The implementation used Python 3.11 and PyTorch 1.10.0 with CUDA 11.3.
- **Model parameters:** Approximately 52M parameters in total (46.54M from the main dual-stream network and 5.5M from the auxiliary SimpleEncoder).
- **Code availability:** `https://github.com/xfJin-tj/AeroVision_PBVS2026`

3.2. ResEffiNet

The method proposed by **Songyq** adopts a dual-stream architecture, as shown in Fig. 3, to leverage complementary information from EO and SAR imagery during training while maintaining SAR-based recognition as the final objective. Two independent backbones are used to learn modality-specific representations without parameter sharing. The EO branch employs a ResNet50 network, while the SAR branch is based on EfficientNet-B0. Through this design, the framework exploits the complementary characteristics of EO and SAR data while preserving SAR-based recognition as the primary inference objective.

3.2.1. Cross-modal feature alignment

To encourage consistency between EO and SAR representations, the framework aligns high-level features extracted from both modality-specific branches during training. Paired EO–SAR samples are processed simultaneously and their corresponding feature representations are compared in order to reduce the discrepancy between the two feature spaces. This alignment mechanism guides the SAR encoder to learn more discriminative representations by leveraging the structural information available in EO imagery.

The training procedure updates both branches while maintaining modality-specific learning dynamics. In particular, an Exponential Moving Average (EMA) teacher mechanism is applied to stabilize optimization and provide smoother target representations during training. This strategy improves the robustness of cross-modal alignment while preserving the specialization of each backbone.

3.2.2. Classification learning

To address the strong class imbalance in the dataset, the model is trained using a class-weighted Focal loss. This loss function places greater emphasis on difficult or underrepresented samples while reducing the contribution of well-classified examples. In addition, class-aware sampling is used during training to improve the representation of minority categories.

3.2.3. Model ensemble

During inference, predictions from multiple trained SAR models are combined to improve robustness. Specifically, the SAR models are loaded from their corresponding checkpoints and evaluated on the test set to produce class logits for each sample. To ensure comparable scales across models, the logits of each model are first normalized using a per-class z-score normalization. The normalized outputs are then fused using a weighted averaging scheme with a ratio of $0.82 : 0.18$ between the two models. The final prediction and confidence score are obtained from the fused logits.

3.2.4. Hardware and Software Specifications

- **Hardware:** Experiments were conducted on a 64-bit Linux workstation equipped with one vGPU-32GB (32 GB VRAM), a 12-core Intel Xeon Platinum 8352V CPU @ 2.10GHz, and 1.0 TiB of RAM.
- **Software:** The framework was implemented in Python 3.8 using PyTorch 2.0.0 and torchvision. Additional li-

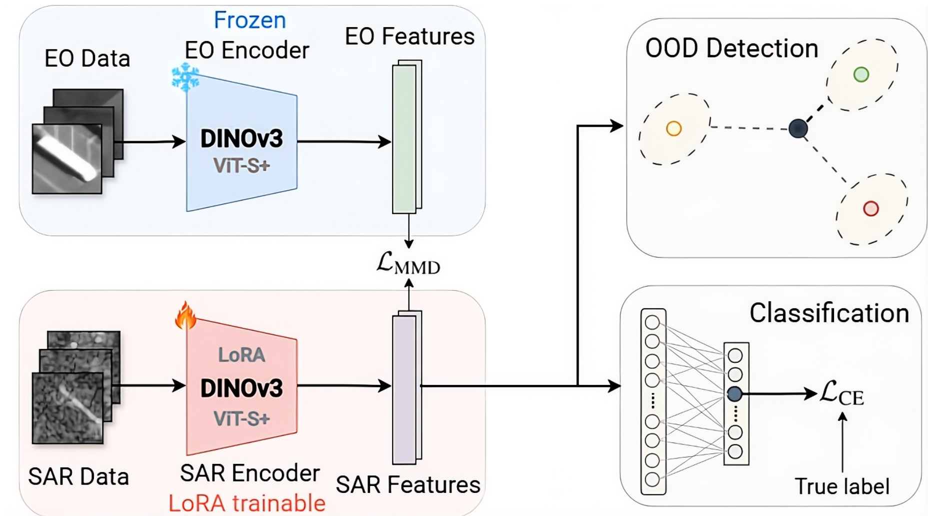

Figure 4. DINOv3 cross-modal alignment approach proposed by IDCOM team.

braries include scikit-learn for data splitting, NumPy and SciPy for numerical operations.
- **Model parameters:** Not explicitly reported.
- **Code availability:** `https://github.com/syqsyq666/MAVIC2026.git`

3.3. DINOv3-based cross-modal alignment

The team by **IDCOM** proposes a cross-modal representation learning approach aimed to improve SAR image classification by leveraging the strong feature extraction capabilities of foundation models trained on optical imagery. As shown in Fig. 4, the method is built on a dual-stream architecture based on the DINOv3 [30] foundation model, where EO and SAR images are processed by two separate encoders. The EO stream uses a frozen DINOv3 Vision Transformer (ViT)-S+ backbone and serves as a reference feature extractor during training. The SAR stream is implemented with a second DINOv3 ViT-S+ model, which is adapted to the SAR domain through Low-Rank Adaptation (LoRA) [11]. This design allows the SAR encoder to benefit from the strong image representation capabilities of the EO branch while preserving a modality-specific learning process.

3.3.1. Feature alignment and classification

To improve the discriminability of SAR representations, the SAR feature space is aligned with the EO feature space during training. This is achieved through a MMD loss applied between the normalized embeddings of both streams. In parallel, classification is performed on the SAR features using a linear classifier trained with a standard cross-entropy loss. The final training objective combines both terms as shown in Eq. (8):

$$L = (1 - \lambda)L_{CE} + \lambda L_{MMD}, \tag{8}$$

where $\lambda \in (0, 1)$ controls the trade-off between classification and cross-modal alignment.

3.3.2. Parameter-efficient SAR adaptation

Rather than fully fine-tuning the SAR backbone, the method employs Low-Rank Adaptation (LoRA) adapters to reduce the number of trainable parameters. Using LoRA, the original backbone weights remain frozen and only a small subset of injected parameters is updated during training. According to the report, this corresponds to approximately 3% of the total model parameters, providing a computationally efficient adaptation mechanism for SAR imagery.

3.3.3. Decoupled OOD detection

A central component of the method is the explicit decoupling of classification and out-of-distribution detection. Instead of using classifier logits or softmax probabilities as confidence scores, OOD detection is performed in the SAR feature space using a distance-based centroid method. Class centroids are computed by averaging training feature vectors for each class, and the confidence score for a test sample is obtained from its minimum Mahalanobis distance [21] to these centroids. The covariance matrix required for this computation is estimated using the Ledoit–Wolf estimator [15]. This OOD module is applied only at inference time and is not included in the training objective.

3.3.4. Hardware and software specifications

- **Hardware:** Experiments were conducted using two NVIDIA GeForce GTX 1080 Ti GPUs (11 GB VRAM each). The total training time was approximately 2 hours.
- **Software:** The implementation used Python 3.13 with PyTorch 2.6 (CUDA 12.6) and the `peft` library (v0.18) for LoRA fine-tuning.

Table 3. Out-of-distribution detection results.

Team	TNR@TPR95 ↑
AeroVision	0.2185
Songyq	0.2290
IDCOM	**0.2695**

Table 4. In-distribution SAR classification results.

Team	AUC ↑	Acc@1 ↑	Final score ↑
AeroVision	0.7690	**0.3005**	**0.4176**
Songyq	0.7687	0.2895	0.4093
IDCOM	**0.7785**	0.2480	0.3806

- **Model parameters:** The model contains approximately 29M parameters at inference time. Due to the use of LoRA adapters, only about 1M parameters (3.54%) were updated during training.
- **Code availability:** https://github.com/luhirsch/mavic-2026-solution

4. Results and analysis

Table 3 shows that the strongest OOD performance is achieved by IDCOM, which attains the highest TNR@TPR95 at 0.2695. This result is consistent with the design of the method that, unlike the other approaches, it explicitly decouples classification from OOD detection and computes OOD scores in the SAR feature space using a Mahalanobis-distance criterion with class centroids. This is a more appropriate mechanism for OOD discrimination than relying on classifier confidence alone, since it evaluates how far a sample lies from the learned IDD structure rather than how strongly it activates a class.

By contrast, AeroVision and ResEffiNet obtain lower TNR@TPR95 values. In both cases, the methods are primarily optimized to improve discriminative classification through cross-modal transfer, class-balancing strategies, and auxiliary training objectives, but without a comparably specialized OOD mechanism. As a result, the learned representations are effective for separating the known target classes, yet less effective at rejecting unseen categories under a strict operating point such as TNR@TPR95. The slightly better OOD result of ResEffiNet (0.2290) over AeroVision (0.2185) may be explained by its simpler SAR-focused inference pipeline, whereas AeroVision prioritizes classification enhancement through auxiliary collaboration and multi-objective optimization, which likely benefits IDD recognition more than boundary formation against unknown samples.

Regarding SAR classification, Table 4 shows that AeroVision achieves the highest total score and therefore ranks first overall. This result is primarily driven by its superior Acc@1, while maintaining an AUC that remains competitive with the other methods. Given that the final score assigns a substantially larger weight to classification accuracy than to OOD performance, AeroVision benefits directly from a method design that prioritizes IDD recognition. Its combination of dual-stream cross-modal adaptation, auxiliary SAR encoding, knowledge distillation, and entity-balanced sampling appears to improve class discrimination more effectively than the other approaches, particularly under the severe class imbalance of the dataset.

ResEffiNet attains the second-best total score, showing a more balanced behavior with competitive classification accuracy and similar AUC, although without surpassing AeroVision in either category. Its simpler dual-stream design, feature alignment strategy, and ensemble-based inference provide a solid improvement over a single SAR classifier, but the method appears less expressive than AeroVision's more elaborate collaborative framework.

In contrast, IDCOM obtains the highest AUC yet ranks third overall because its classification accuracy is substantially lower. This behavior is consistent with its method design, as the DINOv3-based representation learning and decoupled Mahalanobis-based OOD scoring are highly effective for separating IDD from OOD samples, but the corresponding SAR classifier is less competitive on closed-set recognition. As a result, IDCOM excels in uncertainty estimation, whereas AeroVision provides the best overall tradeoff under the scoring rule adopted in the challenge.

5. Identified limitations

Despite the progress demonstrated by the submitted approaches, the results reveal several limitations in current SAR-based recognition methods. Although cross-modal learning strategies successfully improve feature representations by transferring structural information from EO imagery to SAR data, the overall classification performance remains relatively modest. Even the top-performing method achieves an Acc@1 close to 30%, indicating that low-resolution SAR imagery, strong speckle noise, and severe class imbalance continue to pose significant challenges for reliable target discrimination. These results suggest that current architectures and training strategies are still limited in their ability to extract highly separable representations from SAR data under these conditions.

A second limitation is the robustness of OOD detection. While some approaches incorporate dedicated mechanisms for uncertainty estimation, such as distance-based scoring in feature space, the achieved TNR@TPR95 values remain relatively low across all methods. This indicates that reliably distinguishing unseen targets from known classes remains difficult in this setting. In particular, methods that prioritize classification performance often rely on confidence scores derived from classifiers, which are known to produce overconfident predictions. These observations highlight the need for stronger feature representations, improved uncertainty estimation mechanisms, and training strategies that better balance classification accuracy with robust OOD detection.

6. Conclusions

This document reports the results of the 5th Multi-modal Aerial View Imagery Challenge: Classification (MAVIC-C). It focuses on SAR-based aerial target classification while also evaluating the ability of models to detect out-of-distribution (OOD) samples. The challenge is built on the UNICORN V2 dataset, which provides spatially aligned EO and SAR imagery to facilitate the study of cross-modal learning strategies for improving SAR recognition. Model performance is evaluated through a protocol that measures in-distribution classification using Top-1 Accuracy (Acc@1) and OOD detection using the Area Under the ROC Curve (AUC) and TNR@TPR95, with the final ranking determined by a weighted score that prioritizes classification accuracy while still accounting for OOD discrimination capability.

The methods explored in this edition highlight the effectiveness of cross-modal learning strategies that leverage EO imagery to improve SAR feature representations. Most approaches rely on dual-stream architectures and feature-space alignment to transfer structural information from EO to SAR, often combined with auxiliary learning or ensemble strategies to improve robustness under class imbalance. Among the evaluated methods, the DINOv3-based approach achieves the best OOD detection performance with a TNR@TPR95 of 0.2695, demonstrating the benefit of decoupling classification and OOD detection through a Mahalanobis-distance scoring mechanism. In contrast, the AeroVision framework obtains the highest overall score due to its stronger classification accuracy, supported by its cross-modal domain adaptation pipeline and auxiliary SAR encoder. ResEffiNet achieves the second-best overall performance, providing a competitive balance between classification accuracy and OOD detection through a simpler dual-stream design and ensemble-based inference.

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References

[1] N. Anusha and B. Bharathi. Flood detection and flood mapping using multi-temporal synthetic aperture radar and optical data. *Egypt. J. Remote Sens. Space Sci.*, 23(2):207–219, 2020. 1

[2] Reza Mohammadi Asiyabi, Arsalan Ghorbanian, Shaahin Nazarpour Tameh, Meisam Amani, Shuanggen Jin, and Ali Mohammadzadeh. Synthetic aperture radar (sar) for ocean: A review. *IEEE JSTARS*, 16:9106–9138, 2023. 1

[3] Yifang Ban, Puzhao Zhang, Andrea Nascetti, Alexandre R. Bevington, and Michael A. Wulder. Near real-time wildfire progression monitoring with sentinel-1 sar time series and deep learning. *Sci. Rep.*, 10(1):1322, 2020. 1

[4] Abdelmalek Bouguettaya, Hafed Zarzour, Ahmed Kechida, and Amine Mohammed Taberkit. Deep learning techniques to classify agricultural crops through uav imagery: a review. *Neural Comput. Appl.*, 34:9511–9536, 2022. 1

[5] Si-Bao Chen, Qing-Song Wei, Wen-Zhong Wang, Jin Tang, Bin Luo, and Zu-Yuan Wang. Remote sensing scene classification via multi-branch local attention network. *IEEE TIP*, 31:99–109, 2022. 1

[6] Gong Cheng, Xingxing Xie, Junwei Han, Lei Guo, and Gui-Song Xia. Remote sensing image scene classification meets deep learning: Challenges, methods, benchmarks, and opportunities. *IEEE JSTARS*, 13:3735–3756, 2020. 1

[7] Jian Ding, Nan Xue, Gui-Song Xia, Xiang Bai, Wen Yang, Michael Ying Yang, Serge Belongie, Jiebo Luo, Mihai Datcu, Marcello Pelillo, and Liangpei Zhang. Object detection in aerial images: A large-scale benchmark and challenges. *IEEE TPAMI*, 44(11):7778–7796, 2022. 1

[8] Sijia Feng, Kefeng Ji, Fulai Wang, Linbin Zhang, Xiaojie Ma, and Gangyao Kuang. Pan: Part attention network integrating electromagnetic characteristics for interpretable sar vehicle target recognition. *IEEE TGSR*, 61:1–17, 2023. 2

[9] Shijie Hao, Yuan Zhou, and Yanrong Guo. A brief survey on semantic segmentation with deep learning. *Neurocomputing*, 406:302–321, 2020. 1

[10] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *CVPR*, 2015. 4

[11] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models. In *ICLR*, 2021. 6

[12] Yuansheng Hua, Lichao Mou, and Xiao Xiang Zhu. Relation network for multilabel aerial image classification. *IEEE TGSR*, 58(7):4558–4572, 2020. 1

[13] Nathan Inkawhich, Claire Thorp, Justice Wheelwright, Oliver Nina, Dylan Bowald, Angel Sappa, and Erik Blasch. 4th multi-modal aerial view image challenge: Sar classification - pbvs 2025. In *CVPRW*, 2025. 2

[14] Harish C. Kumawat, Mainak Chakraborty, A. Arockia Bazil Raj, and Sunita Vikrant Dhavale. Diat-μsat: Small

aerial targets' micro-doppler signatures and their classification using cnn. *IEEE GRSL*, 19:1–5, 2022. 1

[15] Olivier Ledoit and Michael Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *J. Multivar. Anal.*, 88(2):365–411, 2004. 6

[16] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. Focal loss for dense object detection. In *ICCV*, 2018. 5

[17] Spencer Low, Oliver Nina, Angel D. Sappa, Erik Blasch, and Nathan Inkawhich. Multi-modal aerial view object classification challenge results - pbvs 2023. In *CVPRW*, 2023. 2

[18] Spencer Low, Oliver Nina, Dylan Bowald, Angel D. Sappa, Nathan Inkawhich, and Peter Bruns. Multi-modal aerial view image challenge: Sar classification. In *CVPRW*, 2024. 2

[19] Spencer Low, Oliver Nina, Dylan Bowald, Angel D. Sappa, Nathan Inkawhich, and Peter Bruns. Multi-modal aerial view image challenge: Sar classification. In *CVPRW*, 2024. 2

[20] Ailong Ma, Yuting Wan, Yanfei Zhong, Junjue Wang, and Liangpei Zhang. Scenenet: Remote sensing scene classification deep learning network using multi-objective neural evolution architecture search. *ISPRS J. Photogramm. Remote Sens.*, 172:171–188, 2021. 1

[21] Maximilian Müller and Matthias Hein. Mahalanobis++: Improving OOD detection via feature normalization. In *ICML*, 2025. 6

[22] N. Nagarani, P. Venkatakrishnan, and N. Balaji. Unmanned aerial vehicle's runway landing system with efficient target detection by using morphological fusion for military surveillance system. *Comput. Commun.*, 151:463–472, 2020. 1

[23] Saurabh Vijay Parhad, Krishna K Warhade, and Sanjay S Shitole. Speckle noise reduction in sar images using improved filtering and supervised classification. *Multimed. Tools Appl.*, 83(18):54615–54636, 2023. 1

[24] Alicia Passah, Samarendra Nath Sur, Babusena Paul, and Debdatta Kandar. Sar image classification: A comprehensive study and analysis. *IEEE Access*, 10:20385–20399, 2022. 2

[25] Yalong Pi, Nipun D. Nath, and Amir H. Behzadan. Convolutional neural networks for object detection in aerial imagery for disaster response and recovery. *Adv. Eng. Inform.*, 43: 101009, 2020. 1

[26] Mohammadreza Pourfard, Tahmineh Hosseinian, Roghayeh Saeidi, Sayed Ahmad Motamedi, Mohammad Javad Abdollahifard, Reza Mansoori, and Reza Safabakhsh. Kaze-sar: Sar image registration using kaze detector and modified surf descriptor for tackling speckle noise. *IEEE TGRS*, 60:1–12, 2022. 1

[27] Leo Thomas Ramos and Angel D. Sappa. Multispectral semantic segmentation for land cover classification: An overview. *IEEE JSTARS*, 17:14295–14336, 2024. 1

[28] Leo Thomas Ramos, Edmundo Casas, Cristian Romero, and Francklin Rivas-Echeverría. K-vehicles: A remote sensing dataset for vehicle detection in aerial imagery. In *WACVW*, 2026. 1

[29] Jacob Shermeyer, Daniel Hogan, Jason Brown, Adam Van Etten, Nicholas Weir, Fabio Pacifici, Ronny Hansch, Alexei Bastidas, Scott Soenen, Todd Bacastow, and Ryan Lewis. Spacenet 6: Multi-sensor all weather mapping dataset. In *CVPRW*, 2020. 1

[30] Oriane Siméoni, Huy V Vo, Maximilian Seitzer, Federico Baldassarre, Maxime Oquab, Cijo Jose, Vasil Khalidov, Marc Szafraniec, Seungeun Yi, Michaël Ramamonjisoa, et al. Dinov3. *arXiv preprint arXiv:2508.10104*, 2025. 6

[31] Huiming Sun, Yuewei Lin, Qin Zou, Shaoyue Song, Jianwu Fang, and Hongkai Yu. Convolutional neural networks based remote sensing scene classification under clear and cloudy environments. In *ICCVW*, 2021. 1

[32] Mingxing Tan and Quoc Le. Efficientnet: Rethinking model scaling for convolutional neural networks. In *ICML*, 2019. 4

[33] Cédric Villani. The wasserstein distances. In *Optimal Transport: Old and New*, 2009. 5

[34] Chenchen Wang, Hong Gu, and Weimin Su. Sar image classification using contrastive learning and pseudo-labels with limited data. *IEEE GRSL*, 19:1–5, 2022. 2

[35] Jinwang Wang, Wen Yang, Haowen Guo, Ruixiang Zhang, and Gui-Song Xia. Tiny object detection in aerial images. In *ICPR*, 2021. 1

[36] Junjie Wang, Wei Li, Mengmeng Zhang, Ran Tao, and Jocelyn Chanussot. Remote-sensing scene classification via multistage self-guided separation network. *IEEE TGRS*, 61: 1–12, 2023. 1

[37] Chang Xu, Jinwang Wang, Wen Yang, and Lei Yu. Dot distance for tiny object detection in aerial images. In *CVPRW*, 2021. 1

[38] Yan Zhao, Lingjun Zhao, Siqian Zhang, Kefeng Ji, Gangyao Kuang, and Li Liu. Azimuth-aware subspace classifier for few-shot class-incremental sar atr. *IEEE TGRS*, 62:1–20, 2024. 2